

Computed, recorded, ignored: a self-audit of a generative drug-discovery campaign whose failure was visible in its own output

Obi Ebuka David (Autogon Inc.)

2026-09-05

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Computed, recorded, ignored

A self-audit of a generative drug-discovery campaign whose failure was visible in its own output

Obi Ebuka David, Autogon Inc. Correspondence: davidobi023@gmail.com

Draft v3, 2026-09-05. All references verified against publisher records (REFERENCES.md).

Abstract

We report a self-audit of a generative small-molecule campaign against KRAS in pancreatic ductal adenocarcinoma, conducted by the authors of the generative system itself. The system (ref. 13) produced 73 candidate molecules; three were sent to synthetic chemists and approximately one year of laboratory work produced no compound. The audit finds that the failure was not a failure of prediction. The campaign’s own output file contains a column named **Epoxide Ring Present**, and its value is **True for all three molecules sent for synthesis**. The same file records predicted mutagenicity (AMES) at a median of 0.93 across all 73 molecules, drug-likeness (QED) at 0.39, and synthetic accessibility at 5.47 against 4.21 and 3.84 for the two reference drugs. Every signal required to halt the campaign was computed, written down, and shipped. None of it was connected to a decision. Selection used docking thresholds alone, and the one criterion with authority over the outcome, **Good Docking Quality Overall**, reads True for 72 of 73 molecules. We further show, against a control of 240,214 molecules released by eight published generative models, that this is **not** a general property of generative chemistry: those models produce molecules passing an unmodified published filter at 81 to 89 percent, against 6.8 percent for this campaign. The distinguishing feature was not the generator but the absence of any gate between prediction and purchase order. We rebuild the pipeline so that every stage must reproduce known answers before its output is used, report four validated targets and three completed screens, and document the two occasions on which

our own replacement methods failed in the same manner and were caught by those gates. Code, data, and all negative results are released.

1. Introduction

Computational drug-discovery pipelines are assembled from components that are individually validated and collectively unaudited. A generative model proposes molecules, a docking method scores them, property predictors annotate them, and a shortlist emerges. Each component has a literature establishing what it does well. Very little establishes what happens when they are chained and the chain is trusted.

The limitations of the individual components are known. Generative models propose molecules that cannot readily be synthesised (ref. 1). Structural-alert catalogues over-reject, and Capuzzi et al. report that **87 small-molecule FDA-approved drugs carry PAINS alerts**, cautioning explicitly against using such alerts to triage compounds (refs. 2, 3). Deep-learning docking methods produce physically invalid poses (ref. 4). We claim novelty for none of these.

What this paper contributes is a chain with **ground truth at the end of it**. The molecules did not stop at a benchmark. Three went to synthetic chemists, and the synthesis did not succeed. That allows a question the literature rarely gets to ask: when a computational campaign fails at the bench, was the information needed to prevent it absent, or present and unused?

For this campaign the answer is unambiguous and is the paper’s central finding. The information was present, in the campaign’s own spreadsheet, in a column named after the functional group that broke the chemistry.

We are the authors of the generative system audited here (ref. 13). This is a self-audit, and we regard that as a strength of the evidence rather than a limitation of it: we hold the primary artefacts, including the failure.

2. Provenance of the material audited

```
Medgnosis GenAI system (published, ref. 13)
|
v
KRAS campaign: 73 molecules generated
|
v
Docking: DiffDock + GNINA, 4 KRAS variants
|
v
Property prediction: ADMET-AI
PAINS | AMES | QED | SA score | Epoxide Ring Present
|
| ..... computed, recorded,
|           NOT wired to any decision
v
SELECTION RULE: DiffDock conf >= -1.5
                GNINA affinity <= -5.0 kcal/mol
|
v
Shortlist recorded (73 molecules + 2 reference drugs)
|
v
3 molecules sent to synthetic chemists
|
v
~1 year, no compound delivered
```

Figure 1. The chain, and where it breaks. Property prediction (D) fed the record but not the decision (E). The dotted edge is the paper’s subject.

The generative system and its selection rule are published (ref. 13): molecules were retained on a DiffDock confidence threshold of -1.5 and a GNINA minimised affinity of -5.0 kcal/mol, with ADMET-AI (ref. 12) descriptors computed alongside. No stability or reactivity criterion appears in the method. The campaign output analysed here is the spreadsheet released with this paper, containing 73 generated molecules and two reference drugs, sotorasib (ref. 9) and adagrasib (ref. 10).

3. Results

3.1 The failure was recorded in the file that recommended the molecules

The campaign’s output file carries 32 columns. Among them is **Epoxide Ring Present**.

Table 1. What the campaign computed about the molecules it selected.

quantity recorded	value across 73 generated molecules	reference drugs
Epoxide Ring Present, the three sent to synthesis	True, all three	not applicable
Epoxide Ring Present, all molecules	True for 13 of 73	False
AMES (predicted mutagenicity), median	0.93	not applicable
QED (drug-likeness), median	0.39	typical norm > 0.5
Synthesis Accessibility Score, median	5.47 (22 of 73 above 6.0)	4.21, 3.84
PAINS	flagged 4 of 73	not flagged
Good Docking Quality Overall	True for 72 of 73	not applicable
FDA Approved	False for 73 of 73	not applicable

Three molecules were selected for synthesis: Hit 13, Hit 41 and Hit 73. Each has **Epoxide Ring Present = True**. Epoxides are strained three-membered ethers, opened by nucleophiles, acids and bases, and in these structures they co-occur with hydrazines and free aldehydes that attack them. The audit script identifies twelve molecules whose own functional groups react with one another, including all three synthesis candidates.

The signal was not weak, ambiguous or buried. It was a boolean column named after the offending group, set to True, in the file used to choose what to make.

3.2 The failure is not a property of generative models

If generative models generally produced unmakeable molecules, this campaign would be an instance of a known problem rather than a finding. We tested that directly against the released outputs of eight published generative models, applying **BRENK** and **PAINS** exactly as shipped in **RDKit**, with none of our own modifications.

Table 2. One filter battery, unmodified published catalogues, applied across models.

set	n	pass filter	aromatic ring	median MW
Purchasable compounds (ZINC)	30,000	97.5%	100.0%	523.6
HMM	3,381	89.4%	77.6%	108.1
AAE	27,730	87.7%	97.9%	323.4
VAE	29,297	86.5%	98.3%	306.4
CharRNN	29,653	86.2%	98.2%	310.4
JT-VAE	30,000	86.0%	98.6%	307.8
NGram	7,134	85.5%	89.9%	233.3

set	n	pass filter	aromatic ring	median MW
LatentGAN	26,799	84.7%	97.5%	307.4
Combinatorial	30,000	81.5%	95.4%	325.4
This campaign	73	6.8%	0.0%	226.3

Eight published models, 240,214 molecules, pass at 81 to 89 percent. This campaign passes at 6.8 percent, and **not one of its 73 molecules contains an aromatic ring**, against 95 to 99 percent for every model tested and 100 percent for purchasable compounds. Both reference drugs contain aromatic rings.

The generative literature does not have this problem. This campaign did. The difference lies in what was permitted to stop a molecule, not in what produced it.

A methodological note on this table. The filters are unmodified. Had we applied our own tuned catalogue, the separation would be uninterpretable, because we would have built the instrument that produced it.

3.3 The same unmodified filter rejects both approved drugs

Applying that identical unmodified catalogue to the two reference drugs:

Table 3. The filter that separates the sets also rejects the drugs.

molecule	status	passes unmodified BRENK + PAINS
Sotorasib	approved KRAS G12C inhibitor (ref. 9)	No
Adagrasib	approved KRAS G12C inhibitor (ref. 10)	No

Both fail, on the acrylamide warhead that BRENK flags as `Michael_acceptor_1` (ref. 5). The alert fires on the feature responsible for the drug’s mechanism.

This is the second half of the problem and it points the opposite way. Used as a gate without calibration, the same catalogue that correctly rejects this campaign’s molecules also rejects the two drugs the campaign was trying to emulate. A filter is not a gate until something has established which of its verdicts to believe.

3.4 Three mechanisms of silent failure, five instances

Re-running each component of the vetting stack against molecules whose correct classification is known yields five failures. They reduce to three mechanisms.

Table 4. Failures grouped by mechanism.

mechanism	instance	consequence
M1. Over-broad structural alert	BRENK <code>Michael_acceptor_1</code> on acrylamide	rejects sotorasib and adagrasib
	Nitroso SMARTS <code>[NX3]-[OX2H0,OX1]</code> also matches nitro	rejects venetoclax
	Alkyne rule in our own stability gate	rejects MRTX1133 (ref. 7)
	ADMET <code>DILI < 0.70</code> and <code>hERG < 0.70</code> (ref. 12)	rejects 199 of 200, including both approved drugs
M3. Tool answers a different question	Retrosynthesis (ref. 11) as a feasibility gate	ranks an impossible molecule above a marketed drug

We report three mechanisms rather than five failures deliberately. Grouping is the honest description, and the recurrence of M1 across three independent implementations, two of them widely used and one written by us, is stronger evidence than five unrelated defects would be.

For M3, the numbers: retrosynthesis returns **80 percent of precursors in stock over 6 steps** for an impossible molecule (“Hit 41”, which carries both an epoxide and a hydrazine that opens it) and **71 percent over 6 steps** for adagrasib. A formal disconnection exists for a molecule that decomposes on formation. Retrosynthesis answers “if this existed, what would assemble it” and never “can this exist”.

In every instance the tool returned well-formed output and raised no error.

3.5 The intervention: panels, not references

All five instances are detected by requiring each tool to reproduce known answers before its output is used. The operative variable is panel size.

Table 5. Detection as a function of control-panel size.

control set	size	instances detected
Reference compounds	3	0 of 5
Calibration panel	14	5 of 5

The panel spans the target class, approved covalent inhibitors, and structurally diverse marketed drugs. Cost is one function call per tool invocation. In the rebuilt pipeline the ADMET stage re-runs its control check on every invocation and exits if any control fails.

3.6 The rebuilt pipeline, and what its gates cost

Eleven phases, each with a known-answer gate. **Nine of the eleven gates exist because a tool had already failed silently.**

Table 6. Method validation across four targets.

variant	structure	cognate redock RMSD	control spread	criterion
G12C	8AFB	0.67 Å	2.96 kcal/mol	< 2.0 Å
G12D	9HFK	0.88 Å	4.84 kcal/mol	< 2.0 Å
G12V	9YMQ	1.55 Å	not recorded	< 2.0 Å
G12R	9XB7 (1.36 Å)	0.99 Å	3.13 kcal/mol	< 2.0 Å

The method replaced was blind docking with CNN rescoring, which showed **4.0 kcal/mol run-to-run variance on a single molecule** (sotorasib scored -3.80 and -7.78 on repeat runs) and therefore could not rank.

Target selection by disease prevalence. Among 3,755 PDAC tumours carrying a codon-12 KRAS mutation, the distribution is **G12D 47%, G12V 34%, G12R 17%, G12C 2%** (ref. 8). (*Denominator: codon-12-mutant tumours. As fractions of all PDAC, approximately 40, 29, 14 and 1.7 percent.*) The audited campaign’s docking columns lead with **G12C**, the variant with approved drugs and abundant liganded structures, and the one accounting for 2 percent of this disease. Structure availability selected the target; prevalence did not.

Screens. From 518,662 gated purchasable compounds: G12D $n = 19,639$, G12V $n = 9,913$, covering 81 percent of codon-12-mutant PDAC. Neither produced a compound competitive with the clinical inhibitors. A G12R screen is in progress.

3.7 Our replacement methods failed in the same way, and the gates caught them

Two failures occurred in the pipeline built to prevent failures. Both were detected by control gates, which is the intervention demonstrating itself on its authors.

A ligand scored outside the protein. Our MM-GBSA implementation initially scored a conformer re-embedded from SMILES rather than the docked pose, placing ligand and receptor in unrelated coordinate frames.

The **crystallographic** cognate ligand of 9YMQ scored **+23.44 kcal/mol**, non-binding, and impossible for a ligand resolved within that structure. Scoring the docked pose gives **−33.22 kcal/mol**. The gate that caught it required the cognate ligand to score as a binder.

A gate that did not check what it claimed. An earlier version of that same gate counted returned results rather than testing the ranking its documentation promised, and allowed the broken run to proceed for 37 minutes. A gate that does not check what it claims to check is indistinguishable from no gate. This is the paper’s thesis, applied to the authors, and it is the reason we report it.

Rescoring cannot rank a shortlist flat in the primary score.

Table 7. Two independent rescoring methods on the G12V top-200.

quantity	value
Vina score spread across all 200 compounds	1.80 kcal/mol
Vina’s documented error (ref. 6)	~2.5 kcal/mol
MM-GBSA spread, same compounds	87.28 kcal/mol
Spearman , Vina vs MM-GBSA (n = 200)	+0.106 (p = 0.13)
Spearman , Vina vs Vinardo (n = 200)	+0.259
Shared compounds in the two top-10 rankings	2 of 10

Neither rescoring method is malfunctioning. The shortlist spans less than the primary method’s own error and carries no ordering to recover. Reported without the primary spread, = 0.106 reads as a failed rescoring method rather than a property of the input. Interim estimates during the run were 0.239 (n = 50), 0.347 (n = 60), 0.196 (n = 80) and 0.106 (n = 200); the apparent significance at n = 60 (p = 0.007) was a sampling artefact, and we report it to document how unstable small-n correlations are in this setting.

Search effort buys geometry, not ranking. Every screen ran at docking exhaustiveness 4 while every validation gate ran at 16, so the gate never covered the configuration used.

Table 8. Exhaustiveness sensitivity, 50 compounds, identical embedding seed.

exhaustiveness	mean Δ vs ex128	vs ex128	top-10 retained	median cost
4	+0.08 kcal/mol	0.821	8 of 10	24 s
16	+0.02	0.950	9 of 10	94 s
64	−0.01	0.977	10 of 10	347 s
128	0.00	1.000	10 of 10	747 s

A 31-fold compute increase moves the median score by 0.08 kcal/mol, so the completed screens are sound. The same parameter on G12R moved the cognate’s score only from −9.28 to −9.80 while moving its **pose from 7.19 Å to 0.99 Å**, converting a failed validation into a passed one. Screening consumes rank and tolerates shallow search; validation and pose-dependent rescoring consume geometry and do not.

4. Discussion

The components of this stack behaved as documented. Structural alerts over-rejected, as Capuzzi et al. described (ref. 2). Retrosynthesis did not model stability, which it never claimed to. Docking scores were noisy within their stated error. Nothing malfunctioned.

What failed was the architecture. Property prediction produced a record; selection consulted a threshold. Between the column reporting an epoxide and the decision to synthesise the molecule carrying it, there was no connection. The campaign could see, and could not act on what it saw.

This distinction matters because it changes the remedy. If the problem were prediction quality, the response would be better predictors, and the field is well supplied with those. If the problem is that predictions are

recorded rather than enforced, better predictors change nothing. The remedy is architectural: every stage must be able to stop the pipeline, and must first demonstrate on known answers that it stops the right things.

The benchmark in §3.2 supports this reading. Eight published models produce broadly makeable molecules without any special stability machinery, because they imitate training distributions drawn from real chemistry. This campaign selected against docking thresholds and departed from that distribution, with nothing positioned to notice. **Not one of its 73 molecules has an aromatic ring**, a feature present in essentially every drug-like compound and in both its own reference drugs. A single aromaticity check would have flagged the entire set.

We report our own two failures for the same reason we report the campaign’s. A paper arguing that computational tools fail silently would be self-refuting if it presented its own tools as having worked first time. The relevant difference is that ours were caught, by gates, before anything was ordered.

5. Limitations

One campaign, no control campaign. We cannot separate “the vetting architecture failed” from “this particular campaign was unusual”. The benchmark in §3.2 bounds the second possibility but does not eliminate it.

The synthesis record is not in our possession. The approximately one-year effort is reported from institutional memory. No laboratory notebook, dated protocol or chemist’s report accompanies this paper. Readers should treat the bench outcome as attested rather than documented, and the computational audit, which is fully reproducible, as the evidential core.

n = 73. The audited set is small. All per-molecule claims are stated with counts.

The 14-drug panel is sufficient, not optimal. It detects these five instances. Minimal sufficient panel construction is open.

One target family. All docking results are KRAS.

Self-audit. We built the system audited here. This gives us the primary artefacts and an obvious interest in the interpretation. All data and code are released so the analysis can be repeated against us.

6. Methods

Software. AutoDock Vina 1.2.7 (ref. 6); RDKit 2026.03 (BRENK and PAINS catalogues as shipped); OpenMM 8.6 with openmmforcefields, openff-toolkit and AmberTools (AM1-BCC charges via antechamber), ff14SB with GBn2 implicit solvent; AiZynthFinder with USPTO policy and ZINC stock (ref. 11); ADMET-AI (ref. 12); Meeko for ligand preparation; PDBFixer for receptor preparation.

Structures. 8AFB (G12C), 9HFK (G12D), 9YMQ (G12V), 9XB7 (G12R, 1.36 Å). Intake required a co-crystallised drug-like ligand. Docking boxes were the ligand extent plus 10 Å per dimension.

Filter battery (§3.2). BRENK and PAINS as shipped in RDKit, no exemptions. Generated sets are the released samples of the eight MOSES baseline models, capped at 30,000 molecules each. Molecules failing SMILES parsing are excluded and counted.

Gates. Cognate redock RMSD < 2.0 Å; survival of a 14-drug panel; approved-control retention in ADMET; stability gate before retrosynthesis; cognate ligand scoring as a binder with sensible control ordering in MM-GBSA.

Statistics. Spearman rank correlation throughout, reported with n. Interim estimates are reported alongside final values where they differ materially.

7. Data and code availability

github.com/eobi/pancreatic_cancer_research contains code, gated libraries, screening results, rescoring outputs, the filter battery, and all validation records **including failed validations**. The G12R validation that failed at 7.19 Å is retained alongside the one that passed at 0.99 Å. A record containing only successes cannot support a methodological claim.

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